

homework 1

1.

שאלה 1

קבעו את סדר המד"ר הבאות :

א. $(y'')^4 + 5xy = e^{3x}$

ב. $x^2y' - xy = 2x$

ג. $xy' + y''' = (x - 1)y'' + x$

A. $(y'')^4 + 5xy = e^{3x}$

- **Order: 2** (The highest derivative is the second derivative, y'')

B. $x^2y' - xy = 2x$

- **Order: 1** (The highest derivative is the first derivative, y')

C. $xy' + y''' = (x - 1)y'' + x$

- **Order: 3** (The highest derivative is the third derivative, y''')
-

2.

שאלה 2

העבירו את המד"ר הבאות מכתוב רגיל לכתוב דיפרנציאלי :

א. $yy' = x$

ב. $x^2e^{xy}y' = x\sin(2x) - xye^{xy}$

A.

$$yy' = x$$

Step 1: Replace y' with $\frac{dy}{dx}$.

$$y \frac{dy}{dx} = x$$

Step 2: Multiply both sides by dx .

$$y \, dy = x \, dx$$

Step 3: Rearrange the terms to one side.

$$x \, dx - y \, dy = 0$$

B.

$$x^2 e^{xy} y' = x \sin(2x) - xy e^{xy}$$

Step 1: Replace y' with $\frac{dy}{dx}$.

$$x^2 e^{xy} \frac{dy}{dx} = x \sin(2x) - xy e^{xy}$$

Step 2: Multiply both sides by dx .

$$x^2 e^{xy} \, dy = (x \sin(2x) - xy e^{xy}) \, dx$$

Step 3: Rearrange the terms to one side (subtracting the left side from the right).

$$(x \sin(2x) - xy e^{xy}) \, dx - x^2 e^{xy} \, dy = 0$$

3.

שאלה 3

העבירו את המד"ר הבאות מכתוב דיפרנציאלי לכתוב רגיל:

$$א. \, dy = y \, dx$$

$$ב. \, \frac{1}{y^3} \, dy = \frac{x}{\sqrt{1+x^2}} \, dx$$

A.

$$dy = y \, dx$$

Step 1: Divide both sides by dx to group the differentials.

$$\frac{dy}{dx} = y$$

Step 2: Replace $\frac{dy}{dx}$ with y' .

$$y' = y$$

B.

$$\frac{1}{y^3} dy = \frac{x}{\sqrt{1+x^2}} dx$$

Step 1: Divide both sides by dx to isolate $\frac{dy}{dx}$ on the left side.

$$\frac{1}{y^3} \frac{dy}{dx} = \frac{x}{\sqrt{1+x^2}}$$

Step 2: Replace $\frac{dy}{dx}$ with y' .

$$\frac{1}{y^3} y' = \frac{x}{\sqrt{1+x^2}}$$

4.

שאלה 4

קבעו על כל אחד מהביטויים הבאים האם הוא מהווה פתרון למד"ר.
במידה ולא, פשטו את הביטוי (אם ניתן) כך שיהיה פתרון למד"ר.

א. $y^2 = x^2 + c$

ב. $\ln|y| = 2x + c$

ג. $y = \cos(x) + e^c$

A.

$$y^2 = x^2 + c$$

- **Analysis:** If we differentiate this implicitly, we get $2yy' = 2x$, which simplifies to the ODE $yy' = x$. Integrating that ODE yields exactly $y^2 = x^2 + c$. The constant c is a simple, arbitrary real number with no algebraic restrictions placed upon it.
- **Conclusion: Yes**, it is a valid general solution. No simplification is needed.

B.

$$\ln|y| = 2x + c$$

- **Analysis:** Differentiating this gives $\frac{1}{y}y' = 2$, which is the ODE $y' = 2y$. The constant c can be any real number, so this implicit equation successfully represents the family of solutions.
- **Conclusion: Yes**, it is a valid general solution.
- *(Helpful simplification note: While the current form is valid, it is highly standard practice to solve for y to make it an explicit solution. Doing so gives $|y| = e^{2x+c} = e^c e^{2x}$. By defining a new constant $C = \pm e^c$, we can write the much cleaner explicit solution: $y = Ce^{2x}$).*

C.

$$y = \cos(x) + e^c$$

- **Analysis:** If we differentiate this equation, the constant term e^c disappears, leaving the ODE $y' = -\sin(x)$.

To find the true general solution of this ODE, we integrate: $\int -\sin(x)dx = \cos(x) + C$, where C can be *any* real number (positive, negative, or zero).

However, in the given expression, the term e^c is strictly positive ($e^c > 0$). It is impossible for e^c to represent a negative constant or zero. Because it restricts the family of curves, it is not a complete general solution.

- **Conclusion: No**, it is not a valid general solution in its current form.
- **Simplified Form:** We fix this by replacing the restricted term e^c with a standard, unrestricted constant C . The simplified valid solution is:

$$y = \cos(x) + C$$